Problem 14 | Problems T5_MechanicalProperties.pdf #14 | Brinell hardness

Brinell Hardness of a Steel from the Indentation

Original statement · Problems T5_MechanicalProperties.pdf, p. 5

14. A Brinell indenter of 10 mm in diameter produced an indentation of 2.5 mm in diameter in a steel when a load of 1000 kg was applied. Calculate the Brinell hardness for that material.

Official Solution: 200 HB

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. The indentation is a spherical cap left by a hard ball of diameter D pressed with the load P, and the hardness is $HB = P/S$ [Session 12 Slide 19].
2. P is in kilogram-force (kg in the slide) and the lengths in mm, so HB is in kgf/mm² and is quoted as a number.
3. The test is standard: the load factor $Q = P/D^2$ of the test must be one of those of [Session 12 Slide 20].

Data

Quantity	Symbol	Value
Ball diameter	D	10 mm
Diameter of the indentation	D_i	2.5 mm
Load	P	1000 kg

Unknown: the Brinell hardness HB.

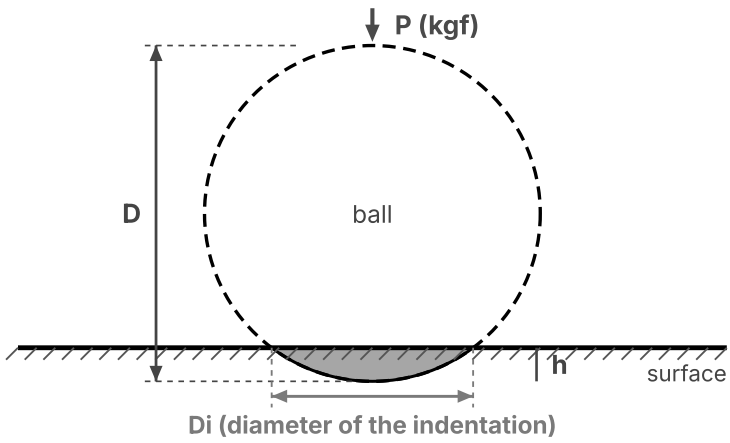
PHASE 2 · FRAMES & STRATEGY

No frames are involved: only one formula, applied directly.

Why this formula. The slide writes $HB = P/S$ with S the area of the spherical cap. For a sphere of diameter D, the cap of height h has area $S = \pi D h$ (Archimedes), and its base circle has radius $D_i/2$, so h follows from $(D/2 - h)^2 + (D_i/2)^2 = (D/2)^2$:

$$h = \frac{D - \sqrt{D^2 - D_i^2}}{2} \quad \Rightarrow \quad S = \frac{\pi D}{2}\left(D - \sqrt{D^2 - D_i^2}\right) \quad \Rightarrow \quad HB = \frac{2P}{\pi D\left(D - \sqrt{D^2 - D_i^2}\right)}$$

which is the expression of [Session 12 Slide 19].



Cap area S = π D h with h = (D − √(D² − Di²)) / 2 ⇒ HB = P / S

Figure 12. Brinell test: the cap of height h left by the ball of diameter D (the proportions are exaggerated, not to scale).

PHASE 3 · STEP-BY-STEP DERIVATION

Square of the indentation diameter and of the ball diameter:

$$D_i^2 = 2.5^2 = 6.25 \text{ mm}^2 \qquad D^2 = 10^2 = 100 \text{ mm}^2$$

Difference and square root:

$$D^2 - D_i^2 = 100 - 6.25 = 93.75 \text{ mm}^2 \qquad \sqrt{D^2 - D_i^2} = 9.68246 \text{ mm}$$

Bracket of the formula and denominator:

$$D - \sqrt{D^2 - D_i^2} = 10 - 9.68246 = 0.31754 \text{ mm} \qquad \pi D \left(D - \sqrt{D^2 - D_i^2} \right) = \pi \times 10 \times 0.31754 = 9.9759 \text{ mm}^2$$

Hardness:

$$HB = \frac{2P}{\pi D \left(D - \sqrt{D^2 - D_i^2} \right)} = \frac{2 \times 1000 \text{ kg}}{9.9759 \text{ mm}^2} = \mathbf{200.5 \text{ HB}} \approx 200 \; HB$$

The depth of the indentation is $h = 0.31754/2 = 0.1588$ mm and the area of the cap $S = \pi D h = 4.988$ mm², so $P/S = 1000/4.988 = 200.5$, the same.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $2P$ in kg (kgf) over an area in mm²: kgf/mm², the unit of hardness (no unit is attached to HB).
- **Load factor.** $Q = P/D^2 = 1000/10^2 = 10$, a value of the table of [Session 12 Slide 20] for $D = 10$ mm (load of 1000 kg), so the test is a standard one.
- **Limit.** For $D_i \ll D$, $D - \sqrt{D^2 - D_i^2} \approx D_i^2/(2D)$ and $HB \approx 4P/(\pi D_i^2) = 203.7$, close to the exact value (the cap is almost flat); for $D_i \to D$, the cap becomes a hemisphere, $S \to \pi D^2/2$, and $HB \to 2P/(\pi D^2)$.
- **Geometry.** $D_i/D = 0.25$: the indentation is a small fraction of the ball, consistent with a hard steel.

Interpretation

The steel has a Brinell hardness of about 200 kgf/mm². It is harder than the plain carbon steel of the table of typical values, 150 HBW [Session 12 Slide 28]. Hardness and tensile strength are both indicators of the plastic deformation, with a relation that depends on the material [Session 12 Slide 30].

Check against the official solution

Item	Official	Computed	Comment
Brinell hardness	200 HB	200.5 HB	agrees